

HSE TOOLBOX TALK

HIGH-PRESSURE WATER JETTING

PURPOSE AND SCOPE

Control injection injury, cutting, recoil, hose-whip and visibility hazards from water jetting through remote methods, a competent team, suitable lance/deadman systems and a full exclusion zone.

KEY HAZARDS

1  INJECTION INJURY	2  CUTTING	3  RECOIL	4  HOSE WHIP
5  FLYING DEBRIS	6  SLIP	7  NOISE	8  LOST VISIBILITY

CRITICAL VALUES TO VERIFY IN THE FIELD

2 Minimum team: operator + controller	100% Deadman and emergency-stop function test	0 Connection work while pressurized
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REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.242(a), 1910.132, 1910.147, manufacturer instructions and WJTA/site procedures integrate equipment condition, PPE, energy control and operational jetting barriers.

OSHA 1910.242(a) SAFE EQUIPMENT

Maintain tools and equipment in safe condition; select pressure components within manufacturer limits and compatibility.

OSHA 1910.132(d) PPE ASSESSMENT

Select face, body, hand, foot, hearing and respiratory protection for jet, product, pressure and debris.

OSHA 1910.147 ENERGY CONTROL

Before approaching nozzle, hose or fittings, stop the pump, isolate energy, bleed pressure and verify.

WJTA / ÜRETİCİ PROSEDÜRÜ DEADMAN AND EXCLUSION ZONE

Use an operator-controlled deadman, hose restraints, suitable lance length, a second controller and complete barricading.

MINIMUM EVIDENCE BEFORE START

- | | |
|---|--|
| ☐ Jetting plan / risk assessment | ☐ Equipment pressure certificates |
| ☐ Hose / fitting inspection | ☐ Deadman / emergency-stop test |
| ☐ Operator competency record | ☐ Barrier and emergency-response plan |

SUPERVISOR DISCUSSION POINTS

- How do we work without anyone entering the jet line?
- What happens if the controller loses sight of the operator?
- What is the first action for an injection injury?

DEADMAN, HOSE RESTRAINT AND FULL EXCLUSION ZONE



Remote method - Deadman - Restraint - Lance - Barrier - Emergency stop

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. CLEAN REMOTELY / AUTOMATICALLY

Remove people from the jet line using robotic, enclosed or remote jetting.

2. CONTAIN THE JET AND HOSE

Restrain hose connections, use protective shrouds and barriers, and control drainage and visibility.

3. DEADMAN AND TWO-PERSON CONTROL

Test deadman, emergency stop and communications; the pump controller must continuously see the operator.

4. JETTING PPE AND EMERGENCY READINESS

Wear cut-resistant suit, visor, boots and hearing protection; treat injection injury as an emergency.

CRITICAL WARNING



A small-looking water-jet wound can cause deep tissue injury and contamination. Isolate pressure and obtain emergency medical evaluation without delay.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|--|---|
| ☐ Method / pressure approved | ☐ Operators competent |
| ☐ Hoses and fittings suitable | ☐ Restraint / shroud fitted |
| ☐ Deadman tested | ☐ Emergency stop tested |
| ☐ Exclusion zone complete | ☐ Drainage and visibility suitable |
| ☐ Medical plan ready | |

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ATTENDANCE AND APPROVAL

Project / Site:

Date:

Presented by:

Work area:

No	Full Name	Company / Role	Signature
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